



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Introduction to Cryptography and Security

Exercise: RSA

Exercise 1

Recall that an RSA public key consists of an RSA modulus N and an exponent e . One might be tempted to use the same RSA modulus in different public keys. For example, Alice might use $(N, 3)$ as her public key while Bob may use $(N, 5)$ as his public key. Alice's secret key is $d_a = 3^{-1} \bmod \varphi(N)$ and Bob's secret key is $d_b = 5^{-1} \bmod \varphi(N)$.

In this question and the next we will show that it is insecure for Alice and Bob to use the same modulus N . In particular, we show that either user can use their secret key to factor N . Alice can use the factorization to compute $\varphi(N)$ and then compute Bob's secret key.

As a first step, show that Alice can use her public key $(N, 3)$ and private key d_a to construct an integer multiple of $\varphi(N)$. Which of the following is an integer multiple of $\varphi(N)$?

① $3d_a$

② $N + d_a$

③ $3d_a - 1$

④ $5d_a - 1$

Exercise 2

Now that Alice has a multiple of $\varphi(N)$ let's see how she can factor $N = pq$. Let x be the given multiple of $\varphi(N)$. Then for any g in $\mathbb{Z}_N^*$ we have $g^x = 1$ in $\mathbb{Z}_N$. Alice chooses a random g in $\mathbb{Z}_N^*$ and computes the sequence $g^x, g^{x/2}, g^{x/4}, g^{x/8} \dots$ in $\mathbb{Z}_N$ and stops as soon as she reaches the first element $y = g^{x/2^i}$ such that $y \neq 1$ (if she gets stuck because the exponent becomes odd, she picks a new random g and tries again). It can be shown that with probability $1/2$ this y satisfies

$$\begin{cases} y = 1 \bmod p, \text{ and} \\ y = -1 \bmod q \end{cases} \quad \text{or} \quad \begin{cases} y = -1 \bmod p, \text{ and} \\ y = 1 \bmod q \end{cases}$$

How can Alice use this y to factor N ?

- | | |
|----------------------------|----------------------------|
| ① compute $\gcd(N - 1, y)$ | ③ compute $\gcd(N + 1, y)$ |
| ② compute $\gcd(N, y - 1)$ | ④ compute $\gcd(N, y^2)$ |

Exercise 3

In standard RSA the modulus N is a product of two distinct primes. Suppose we choose the modulus so that it is a product of three distinct primes, namely $N = pqr$. Given an exponent e relatively prime to $\varphi(N)$ we can derive the secret key as $d = e^{-1} \bmod \varphi(N)$. The public key (N, e) and secret key (N, d) work as before. What is $\varphi(N)$ when N is a product of three distinct primes?

$$\textcircled{1} \quad \varphi(N) = (p-1)(q-1)(r-1)$$

$$\textcircled{2} \quad \varphi(N) = (p-1)(q-1)$$

$$\textcircled{3} \quad \varphi(N) = (p-1)(q-1)r$$

$$\textcircled{4} \quad \varphi(N) = (p-1)(q-1)(r+1)$$

An administrator comes up with the following key management scheme: he generates an RSA modulus N and an element s in $\mathbb{Z}_N^*$. He then gives user number i the secret key $s_i = s^{r_i}$ in $\mathbb{Z}_N$ where r_i is the i 'th prime (i.e. 2 is the first prime, 3 is the second, and so on).

Now, the administrator encrypts a file that is accessible to users i, j and t with the key $k = s^{r_i r_j r_t}$ in $\mathbb{Z}_N$. It is easy to see that each of the three users can compute k . For example, user i computes k as $k = (s_i)^{r_j r_t}$. The administrator hopes that other than users i, j and t , no other user can compute k and access the file.

Unfortunately, this system is terribly insecure. Any two colluding users can combine their secret keys to recover the master secret s and then access all files on the system. Let's see how. Suppose users 1 and 2 collude. Because r_1 and r_2 are distinct primes there are integers a and b such that $ar_1 + br_2 = 1$. Now, users 1 and 2 can compute s from the secret keys s_1 and s_2 as follows:

$$\textcircled{1} \quad s = s_1^a \cdot s_2^b \text{ in } \mathbb{Z}_N.$$

$$\textcircled{2} \quad s = s_1^b \cdot s_2^a \text{ in } \mathbb{Z}_N.$$

$$\textcircled{3} \quad s = s_1^b + s_2^a \text{ in } \mathbb{Z}_N.$$

$$\textcircled{4} \quad s = s_2^a \text{ in } \mathbb{Z}_N.$$



25
SOICT

VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Thank you!

